GSLP-CIM: A 28-nm Globally Systolic and Locally Parallel CNN/Transformer Accelerator With Scalable and Reconfigurable eDRAM Compute-in-Memory Macro for Flexible Dataflow

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***Abstract*— This article reports a globally systolic and locally parallel (GSLP) convolutional NN (CNN) and Transformer accelerator based on the scalable and reconfigurable (SR) embedded dynamic random-access memory (eDRAM) compute-in-memory (CIM) macro. It features: 1) a GSLP architecture employs systolic CIM macros with the reconfigurable inter-CIM network to support flexible dataflow, including weight stationary (WS), output stationary (OS), and Row stationary (RS); 2) an SR-CIM macro features reconfigurable weight/input/output memory ratio to maximize the related data reuse in different dataflow; 3) a high-density 3T eDRAM-CIM cell to further improve the density of the accelerator; 4) an area-efficient in-memory accumulator (IMA) to save the area and power overhead of the digital accumulation in each CIM macro. Prototyped in 28-nm CMOS process, the proposed GSLP-CIM accelerator exhibits a 4b peak throughput density of 0.16 TOPS/mm^2 and a 4b peak compute energy efficiency of 3.55 TOPS/W. Specifically, evaluated with ResNet-50@ImageNet and ViT-B@ImageNet, this work reaches the system throughput of 24.5 and 5.66 inferences per second (IPS), the system throughput density of 19.3 IPS/mm^2 and 4.46 IPS/mm^2, the system compute energy efficiency of 423.9 inferences per watt (IPW) and 97.6 IPW, respectively.**

***Index Terms*— Neural network (NN), transformer, embedded dynamic random-access memory (eDRAM), compute-in-memory (CIM), systolic, flexible dataflow.**

I. Introduction

With the rapid evolution of artificial intelligence (AI), the deep neural networks (DNNs) including convolutional NN (CNN) and Transformer-based large language model (LLM) have been widely employed in various applications such as object detection and classification [1], [2], autonomous driving [3], natural language processing [4] and so on.

Recently, the Transformer-based LLMs incorporating vision Transformer (ViT) and generative pre-trained Transformer (GPT) have obtained more attention by achieving superior accuracy than CNNs, especially in complex AI tasks [4], [5], e.g., the GPT-2 model achieves 20.4% higher accuracy than long short-term memory (LSTM) for language modeling [6]. However, the Transformer-based models need many more parameters and matrix-vector multiplications (MVMs) than CNNs. Compared with ResNet-50 for ImageNet classification, the ViT-B model needs 3.4× more parameters with only 1.7% higher accuracy [7]. Therefore, to efficiently accelerate DNN models with acceptable accuracy among various simple and complex AI tasks, both CNN and Transformer-based LLM should be supported in DNN accelerator design.

Fig. 1(a) depicts the architecture of a classic digital DNN accelerator [8], [9], [10], [11]. Due to the separated memory and compute, there is intensive data movement between the memory and the processing elements array (PEA). Since the data movement dominates the power consumption of the DNN accelerator [32], existing digital accelerators are focused on optimizing the dataflow to maximize the data reuse within PEA. Following the definition of dataflow in Eyeriss [9], the popular dataflow can be typically classified into weight-stationary (WS) [8], [12], input-stationary (IS) [13], output-stationary (OS) [14], and row-stationary (RS) [9] which can maximize the data reuse of weights, inputs, outputs, and all of them in row-wise, respectively. For dealing with different network structures with various matrix dimensions of weight and input/output in CNN models and Transformer-based LLMs, the PEA is required to support flexible dataflow (FD) [15], [26].

Fig. 1(b) shows the compute-in-memory (CIM) approach, which integrates the MVM operations into the memory array to reduce the data movement between memory and compute units. Therefore, the CIM is an energy-efficient scheme for the MVM-intensive DNN accelerator design [16]. Since the WS is an efficient dataflow for accelerating general CNN models [12], in most of existing CIM-based accelerators [8], [17], [18], [19], [20], [21], [26], [27], [28], [31], only weights are stored in the CIM macro and only WS dataflow can be

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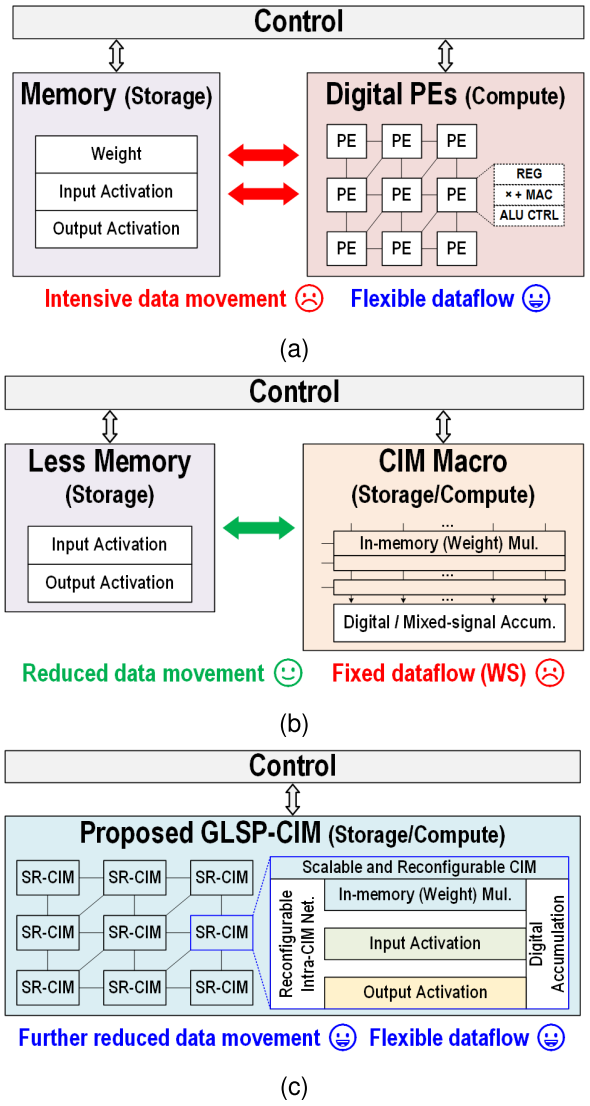


Fig. 1. Architecture of (a) classic digital DNN accelerator, (b) typical CIM-based DNN accelerator with weight stationary, and (c) proposed GSLP-CIM with scalable and reconfigurable CIM macro for flexible dataflow (FD).

supported. However, the data movement of activations between CIM macro and activation memory still exists, and the FD required by the Transformer-based LLMs cannot be supported, which leads to insufficient data reuse in CIM macro and redundant off-core data access. Since the data access from the on-chip global buffer and the off-chip memory consumes ~6× and ~200× larger power [32] than typical in-core memory, respectively, the off-core data access in this work includes both the data access from the on-chip global buffer and the off-chip memory.

Although the FD can be supported in [40], it is only designed for a binary and small CNN model (i.e., AlexNet) with a fixed weight-feature memory ratio. However, recent large and complex emerging AI models (e.g., residual-based CNN and the Transformer-based LLM) require a multi-bit operation and a reconfigurable weight/input/output ratio for sufficient accuracy and minimum data access. Moreover, there is no optimization for scalability in [40]. A systolic CIM array published in [41] shows a potential in scalability. However, only WS dataflow can be supported in [41].

To surmount the challenge raised before, we propose the GSLP-CIM: a globally systolic and locally parallel (GSLP) accelerator based on a scalable and reconfigurable embedded dynamic random-access memory (eDRAM) based CIM macro. The proposed GSLP-CIM combines and further promotes the superiority of FD in digital DNN accelerators and the advantage of reduced data movement in CIM, as illustrated in Fig. 1(c). Specifically, the primary contributions from architecture to circuit are listed as follows.

1) A GSLP architecture which employs a systolic CIM macro array with a reconfigurable inter-CIM communication network. Implementing the FD (WS/OS/RS) within CIM (FD-CIM), achieving scalability, and supporting both CNN and Transformer-based LLM with 4b/8b operation.
2) A scalable and reconfigurable eDRAM-CIM (SR-CIM) macro that stores both weights and input/output activations and can be reconfigured in different weight/input/output memory ratio. By allocating optimized memory ratio under different dataflow, the local data reuse of weight/input/output can be maximized, minimzing the repeated off-core data access.
3) A high-density 3T eDRAM-CIM cell achieves 1.3× smaller cell area than an SRAM-CIM cell [22] to improve the density of the entire accelerator further.
4) An area-efficient in-memory accumulator (IMA) based on a ping-pong eDRAM set to further reduce the area and power overhead of the digital accumulation in CIM macro.

The remainder of this paper is organized as follows. Section II presents the overall architecture of the proposed GSLP-CIM CNN/Transformer accelerator, the structure of SR-CIM macro, and the related design details for FD. Section III introduces the proposed high-density 3T eDRAM-CIM cell and the IMA. Section IV describes the implementation results and the performance comparison. Finally, Section V concludes this work.

II. Architecture of GSLP-CIM and SR-CIM Macro for Flexible Dataflow

A. Overall Architecture of the Proposed GSLP-CIM

Fig. 2 illustrates the overall architecture of the proposed GSLP-CIM CNN/Transformer accelerator. It incorporates a 2×2 SR-CIM macro array, a reconfigurable inter-CIM communication network, and the weight/input/output buffers, which only connect to the edge SR-CIM macros. Globally, the 2×2 SR-CIM macro array is operated in systolic. In each SR-CIM macro, the weights and input/output activations are allocated with different memory, and the MVM operations are executed in parallel within the weight memory.

Based on the reconfigurable inter-CIM network, the GSLP-CIM supports flexible (WS/OS/RS) dataflow, as shown in Fig. 3. The WS dataflow is depicted in Fig. 3(a). In global, the weights are transferred horizontally from west to east, the input activations (IAs) are transferred vertically from south to

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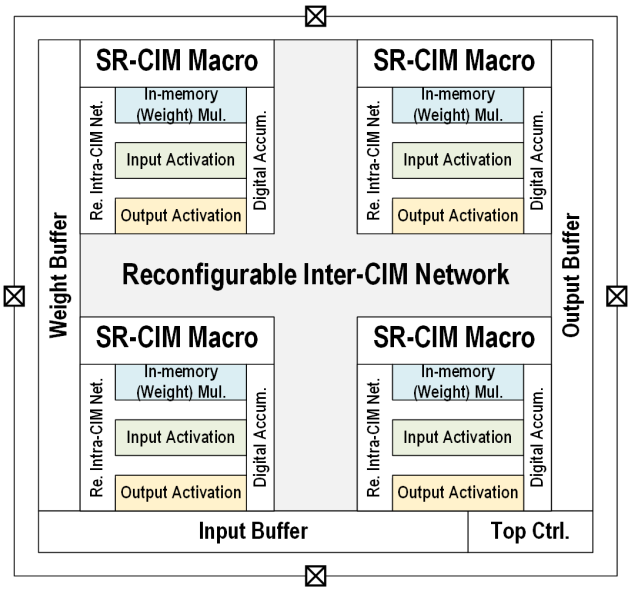


Fig. 2. Overall architecture of the proposed GSLP-CIM.

north, and the output activations (OAs) or partial sums (p-sums) are accumulated horizontally from west to east. At the local level, most of the memory in each SR-CIM macro stores the weights to maximize the reuse of weights.

For the OS dataflow, the global data transfer direction of weights and input/output activations are the same as the WS dataflow. To maximize the reuse of OAs or p-sums, the OAs or p-sums are accumulated locally. Most of the memory in each SR-CIM macro is employed to store the OAs or p-sums, as shown in Fig. 3(b).

Fig. 3(c) illustrates the RS dataflow in GSLP-CIM. Globally, the weights are transferred horizontally from west to east, the IAs are transferred diagonally, and the OAs or p-sums are accumulated vertically from south to north. Locally, each SR-CIM macro stores rows of weights, IAs, and OAs or p-sums to reuse the data row-wisely.

It should be noted that, the horizontal movement of weights in GSLP-CIM does not cause additional data movement. It achieves off-chip data access reduction and scalability. Limited by the I/O bandwidth and memory density, the conventional WS-CIM designs typically have a limited size and hence cannot store an entire layer. Thus, WS-CIM requires massive partial sum data movement. By transferring the weights horizontally, the weights can be reused by neighbor CIM macros to compute with different inputs, enabling FD-CIM to reduce the repeated access of p-sums among various models and tensor shapes. A detailed numerical study can be found in Section II-C. Moreover, it only requires a 1:2 demultiplexer (DEMUX) in each RBL of weight memory to support the horizontal move, consuming $<1\%$ power overhead.

B. Scalable and Reconfigurable eDRAM-CIM (SR-CIM) Macro

Fig. 4 shows the structure of the SR-CIM macro. Each SR-CIM macro includes an 18-kb main eDRAM-CIM array, which is organized in 16 channels with 18 rows and 64 columns, an IA local buffer (IALB) which contains 3×64 eDRAM cells and is used to store the IAs for WS and OS dataflow, a set of proposed area-efficient IMA and related adder tree for bit-wise and Kernel/channel-wise accumulation, respectively. The pre-adder tree accumulation is employed in this work to improve the flexibility of processing different channel sizes in various models.

The write BLs (WBLs) and RBLs in the eDRAM-CIM array are segmented by the BL switches to achieve reconfigurbale weight/input/output ratio and support FD within CIM (FD-CIM), as illustrated in Fig. 5. Different from the resource switching method in [31], whose entire CIM macro can only store weight, input, or output, one at a time, this work can simultaneously store all of them in a macro with reconfigurable ratios. It enables us to support more dataflow types to reduce off-core data access by $>95\%$ compared to the single WS dataflow [31]. Moreover, the CIM cell configured as non-weight storage in [31] will lose its computing function, reducing the overall throughput. In comparison, SR-CIM maintains their computing function regardless of their memory configuration. Fig. 5(a) depicts the eDRAM-CIM array for WS dataflow. There is no segmentation of the eDRAM-CIM array, and all of the 18 eDRAM rows store the weight for maximizing the weight reuse.

For OS dataflow, the eDRAM-CIM array is segmented into 2 parts (i.e., 1 eDRAM row and 17 eDRAM rows): the 1 eDRAM row is employed to store the weight and execute the in-memory multiplication, while the 17 eDRAM rows are used to store the OAs or p-sums, as shown in Fig. 5(b). Hence, the OAs and p-sums can be accumulated and stored in the local SR-CIM macro as much as possible to minimize the movement of OAs or p-sums.

Fig. 5(c) shows the memory allocation for RS dataflow. The eDRAM-CIM array is segmented into 3 parts to store weights, IAs, and OAs or p-sums. For row-wise processing, 3 memory rows in each eDRAM-CIM channel are allocated for storing weights and IAs to match the most popular kernel size of 3×3 and enable the row-wise data reuse locally. Following the principle of maximizing local accumulation in [32], the remaining 12 rows are allocated for the OAs or p-sums to maximize on-chip accumulation and hence reduce the off-core data access of p-sums. The details of the data access reduction will be shown in Section II-C.

The proposed GSLP-CIM accelerator can be programmed layer-wise to achieve optimal dataflow and reduce off-core data access. The BL switches dynamically redistribute the weight/input/output ratio in the eDRAM-CIM array to switch dataflow between different layers.

C. Workload Mapping in GSLP-CIM

Fig. 6 further illustrates the workload mapping of the GSLP-CIM accelerator and the related reconfiguration of the circuits in SR-CIM macro for different dataflows. Fig. 6(a) shows the details of WS dataflow. Globally, the weights are vertically mapped to the SR-CIM macro array Kernel-wise, while horizontally mapped to the SR-CIM macro array channel-wise. For example, in the 2×2 SR-CIM macro array, the south and north macros store Kernels 0-31 and 32-63,

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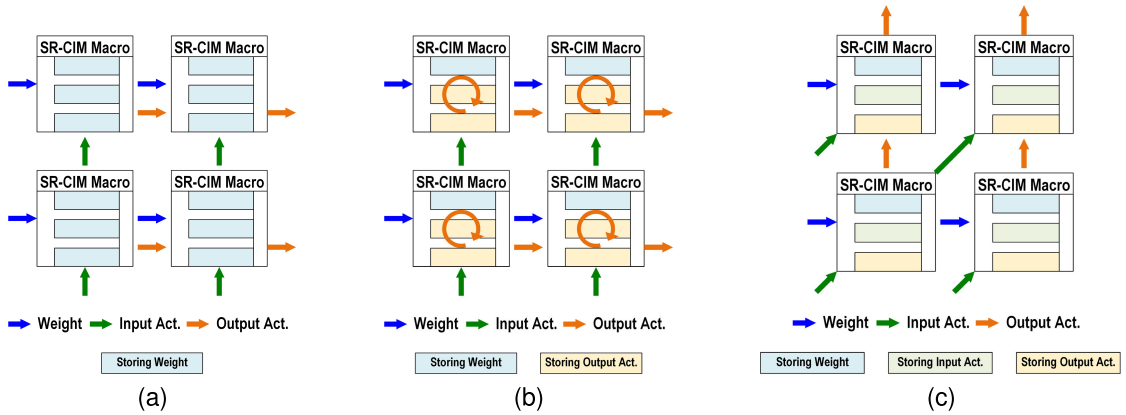


Fig. 3. FD in GSLP-CIM: (a) weight-stationary (WS); (b) output-stationary (OS); (c) row-stationary (RS).

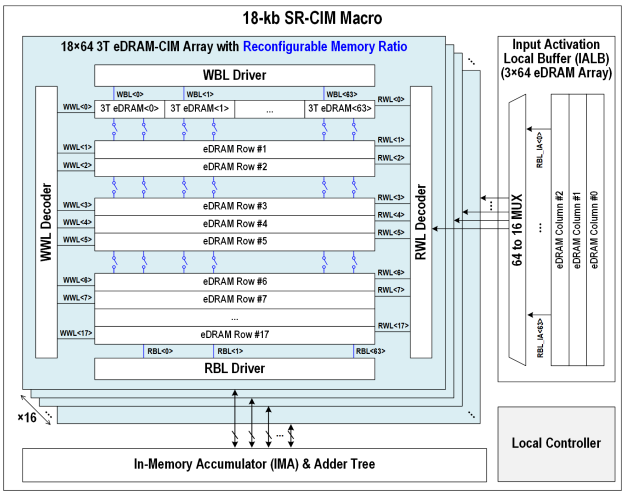


Fig. 4. Structure of the proposed SR-CIM macro.

and the west and east macros store channels 0-15 and 16-31, respectively. Locally, the weights are mapped to each eDRAM-CIM channel kernel-wise, while to different channels of the eDRAM-CIM array channel-wise, as shown in the bottom of Fig. 6(a). The IAs are mapped to each row of the IALB channel-wise for local parallel MVM operations. Taking 4b MVM as an example, the 16 channels 4b IAs are read out to the input of 64-to-16 multiplexer (MUX) and then sent to the 16 eDRAM-CIM channels in 4 cycles, performing bit-serial computation of IAs. Selected by the 1-to-18 MUX in each eDRAM-CIM channel, the IAs are serially transferred to each read word line (RWL) of the eDRAM-CIM row, which stores the 4b weights from 16 Kernels. Therefore, the 16×64.1b MVM can be executed parallel in each SR-CIM macro. For processing 8b MVM, this work follows the bit-parallel and bit-serial (BPBS) scheme in [29]. Towards data mapping, each eDRAM-CIM row stores 8×8b weights, and the IALB employs 2 rows to store the 4b MSBs and 4b LSBs of the IA from 16 channels for ensuring the parallel computation of 16 eDRAM-CIM channels.

Fig. 6(b) depicts the design details of OS dataflow. Globally, the weights are vertically mapped to the SR-CIM macro array Kernel-wise, while they are broadcast horizontally through the SR-CIM macro array. E.g., the south and north macros store Kernel 0-15 and Kernel 16-31, respectively, while both the west and east macros store the same weights. The local weight and IA mapping are the same as the WS dataflow. During the MVM operation, since there is only one eDRAM-CIM row for weights, the 1-to-18 MUXs are bypassed. To match the dataflow of local parallel MVM operation, the OAs and p-sums are accumulated locally and stored channel-wise to the 16 eDRAM-CIM channels.

Fig. 6(c) shows the related details of RS dataflow. Globally, the weights are vertically mapped to the SR-CIM macro array row-wise, while broadcast through the SR-CIM macro array horizontally. For example, the first and second rows of weights from Kernel 0-15 and channel 0-15 are stored in the south and north macros, respectively. When the kernel size is 1×1, the weights are also vertically mapped to the SR-CIM macro array Kernel-wise. Locally, in each eDRAM-CIM channel for weights, the 3 eDRAM-CIM rows store the weights row-wisely, while the 64 eDRAM-CIM columns also store the weights Kernel-wise. The weights are also mapped channel-wise for different channels of the eDRAM-CIM array. The IAs are mapped to each row of the eDRAM-CIM array for IAs channel-wise, while are mapped to the 3 eDRAM-CIM rows and 17 eDRAM-CIM channels row-wise. During the MVM operation, the 1-to-18 MUXs are reconfigured to 1-to-3 MUXs to match the 3 eDRAM-CIM rows for weights. Similarly, the OAs and p-sums are accumulated locally and stored channel-wisely to the 16 eDRAM-CIM channels, while row-wisely to each eDRAM-CIM channel.

Towards the matrix multiplication in Transformer-based models, taking the generation of Query (Q)/Key (K)/Value (V) as an example, the dimension of each token can be mapped to the channels in convolution, the dimension of each output token and head can be mapped to the kernels with a kernel size of 1×1 in convolution, and the input/output tokens can be mapped to the input/output features [38], [39]. For processing the $Q \times K^T$, the K matrix should be fetched to the GSLP-CIM in a transposed way [33]. The WBL driver with a transposable output buffer is employed for the transposed fetching, causing 1.1% power consumption in each SR-CIM macro.

Based on the proposed GSLP-CIM accelerator with FD, the off-core data access, including the on-chip global buffer access and the off-chip access of weights, IAs, and OAs

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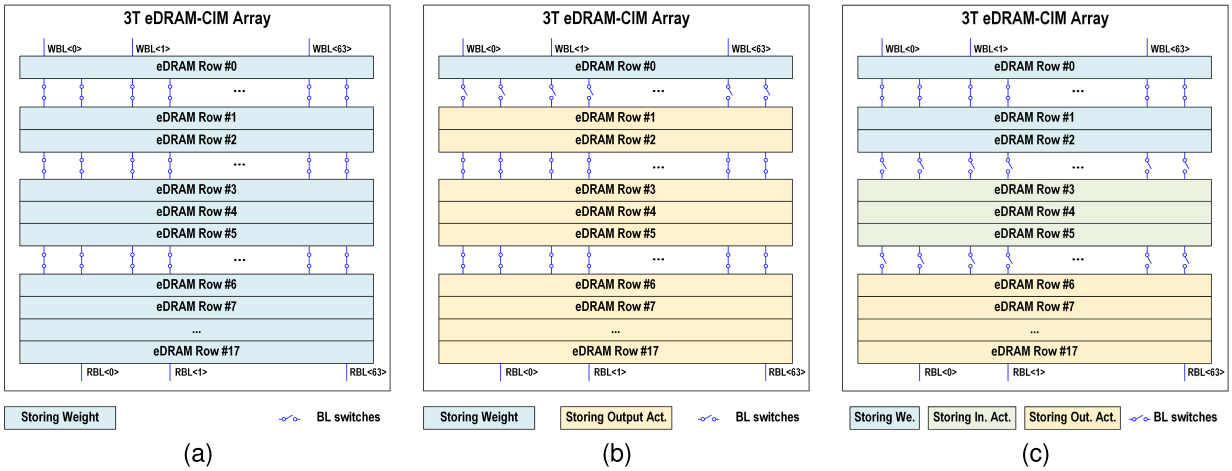


Fig. 5. Reconfigurable weight/input/output memory ratio based on the segmented WBL and RBL in SR-CIM for (a) WS dataflow, (b) OS dataflow, and (c) RS dataflow.

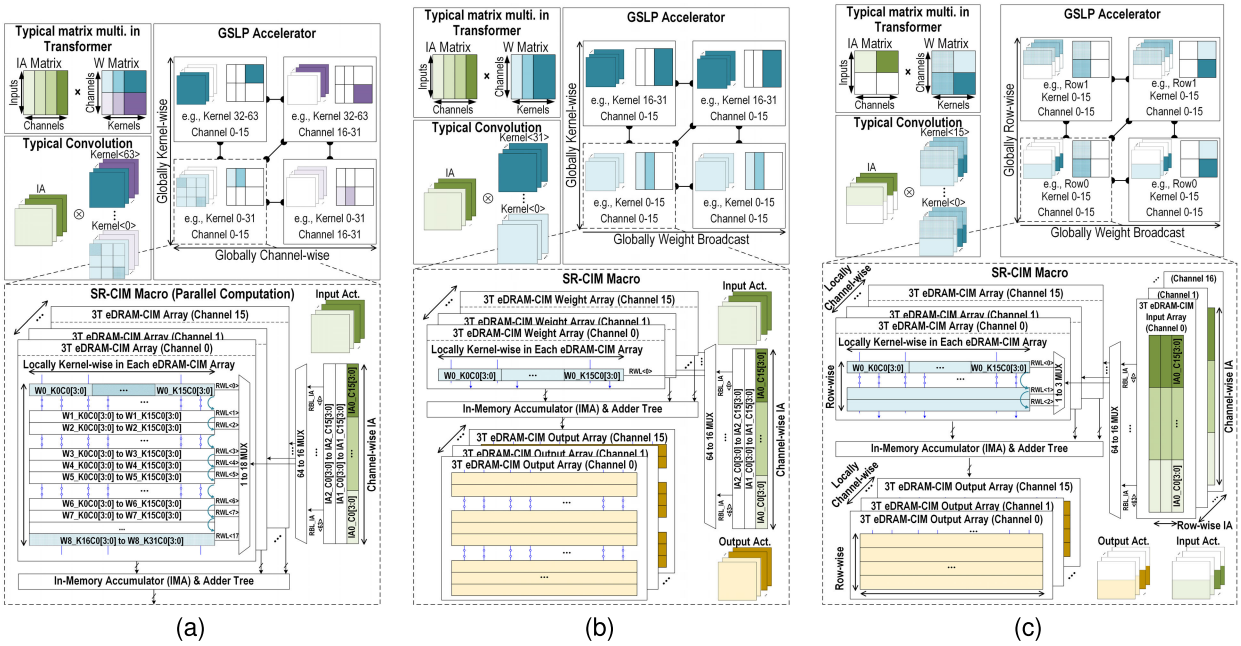


Fig. 6. Workload mapping in GSLP-CIM and the related reconfiguration of the circuits in SR-CIM macro for (a) WS dataflow, (b) OS dataflow, and (c) RS dataflow.

can be significantly reduced than the existing fixed WS-CIM works [17], [18], [19], [20], [21], [26], [27], [31]. Hence, the reduced data access avoids the redundant MVM operations as much as possible. Fig. 7 takes several convolutional layers (C18-to-C30) in the ResNet-50@ImageNet and a layer of ViT-B@ImageNet as examples to demonstrate the advantages. Evaluated on the GSLP-CIM accelerator with 72-kb memory and 4096 × 1b parallel multiplication, and based on the FD and the related workload mapping scheme as illustrated in Fig. 3 and Fig. 6, respectively, the data access amount can be calculated by the layer configuration including the channel number, kernel number, kernel width, kernel height, and output feature size.

As shown in Fig. 7(a), the WS-CIM has the minimum data access in C18, C19, C21, C22, C25, and C28, while the RS-based CIM scheme (RS-CIM) has the minimum data access in C20, C23, C24, C26, C27, C29, and C30. With the channel number and kernel number of 256, kernel width and kernel height of 3, and output feature size of 196, the off-core data access of WS-CIM is ∼348 MB which is 80.4× than the RS-CIM in C24, C27, and C30. Fig. 7(b) shows that the RS-CIM has the minimum data access in the computation of generating Query (Q)/Key (K)/Value (V), the feedforward layer, and the fully connect (FC) layers. With the channel number of 3072, kernel number of 768, kernel width and kernel height of 1, and output feature size of 197, the off-core data access of WS-CIM is ∼474.5 MB which is 79.2× than the RS-CIM for computing the second FC layer.

Fig. 7(c) further shows the normalized execution cycles with different dataflow-based CIM schemes. The WS-CIM has the least execution cycles in C18 and C21, the OS-based CIM scheme (OS-CIM) has the least execution cycles in C23-to-C30, and the RS-CIM has the least execution cycles in C19, C20, and C22. The WS-CIM even has 15.7× more execution cycles than the OS-CIM in C24, C27, and C30. Based on the FD and operated at 400 MHz, the total latency for processing

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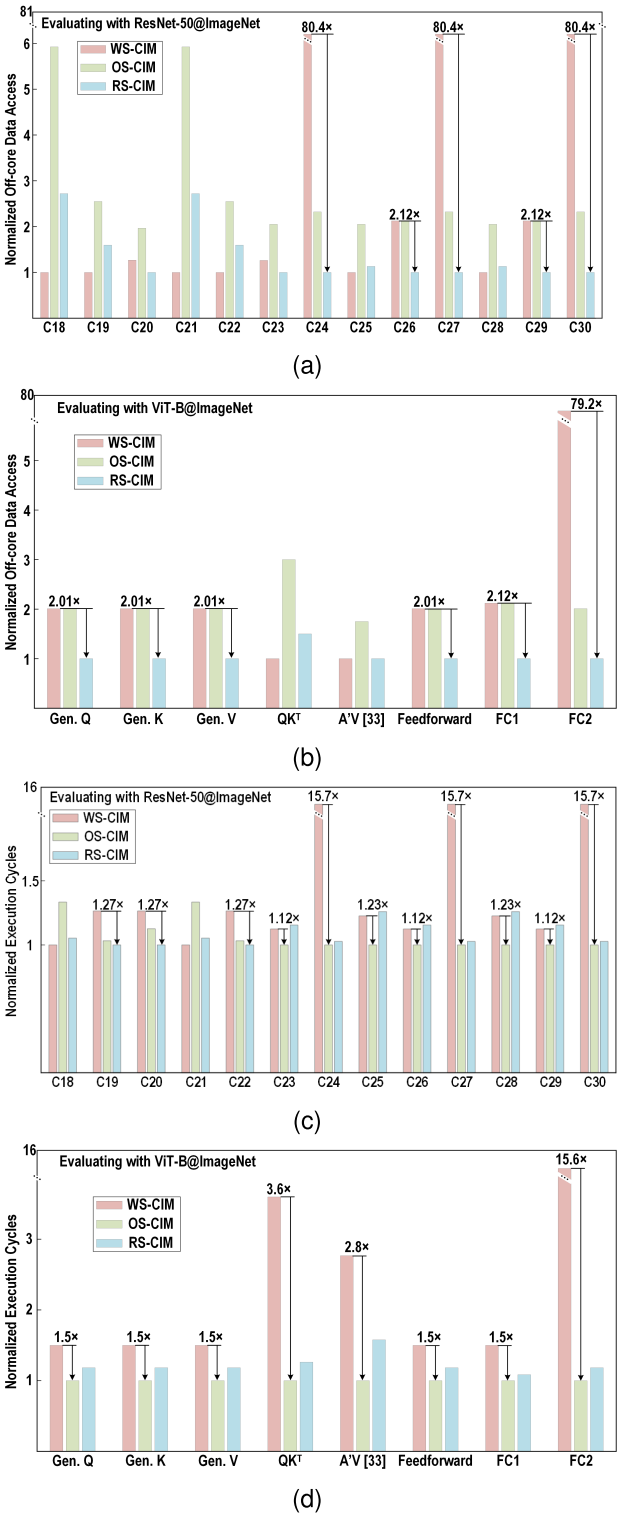


Fig. 7. Normalized off-core data access based on different dataflows for inferencing (a) ResNet-50@ImageNet (takes convolutional layer-18 to layer-30 as an example) and (b) ViT-B@ImageNet, and normalized execution cycles for inferencing (c) ResNet-50@ImageNet and (d) ViT-B@ImageNet.

C2-C49 is ∼42.6 ms, and 9% of the execution cycles are used for off-core data access. Also, as illustrated in Fig. 7(d), the OS-CIM has the least execution cycles for accelerating ViT-B@ImageNet. The WS-CIM even has 15.6× more execution cycles than the OS-CIM for computing the second FC layer. The latency for processing each layer in ViT-B is ∼15.6 ms,

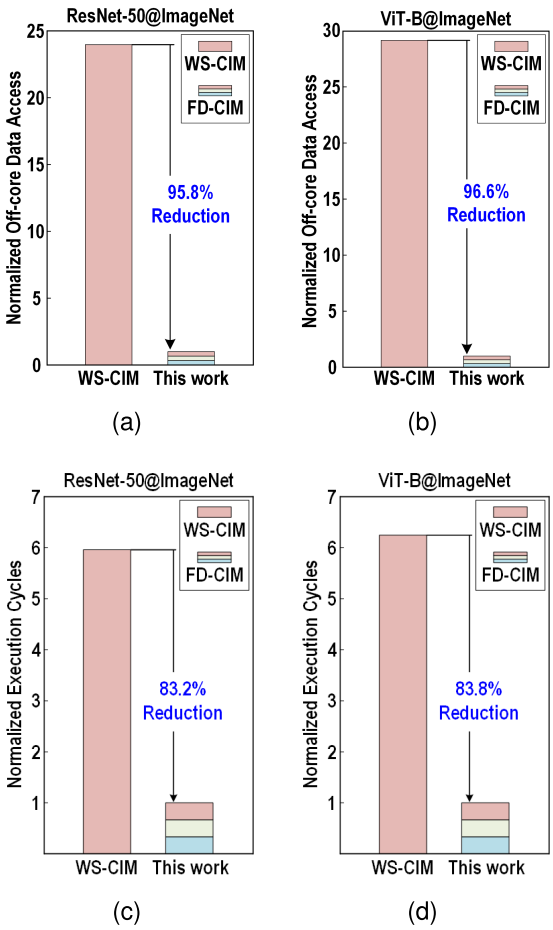


Fig. 8. Normalized off-core data access reduction of GSLP-CIM with FD than WS-CIM counterpart in (a) ResNet-50@ImageNet, and (b) ViT-B@ImageNet; Normalized execution cycles reduction of GSLP-CIM with FD than WS-CIM in (c) ResNet-50@ImageNet, and (d) ViT-B@ImageNet.

and about 8% of the execution cycles are used for the off-core data access. Therefore, for processing different convolutional layers, the single and fixed dataflow-based CIM scheme cannot reach the minimum data access or execution cycles.

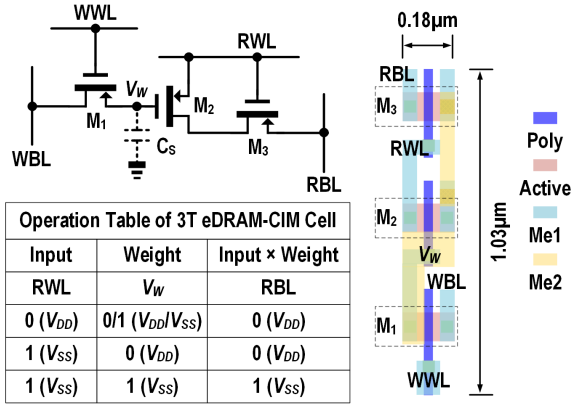
The proposed FD-CIM employs the optimized dataflow in each layer to achieve the minimum data access and execution cycles among the whole model. Evaluated with ResNet-50@ImageNet and ViT-B@ImageNet, the FD-CIM achieves 95.8% and 96.6% off-core data access reduction and saves 83.2% and 83.8% execution cycles than the WS-CIM counterpart [8], [17], [18], [19], [20], [21], [26], [27], [28], as illustrated in Fig. 8.

III. Proposed 3T eDRAM-CIM Cell and In-Memory-Accumulator

A. High-Density 3T eDRAM-CIM Cell

Fig. 9 shows the schematic (left and top), operation table (left and bottom), and layout (right) of the proposed high-density 3T eDRAM-CIM cell. M_1 is the write access transistor which is controlled by the write WL (WWL) to write data from the WBL to the storage node. The data is stored by the parasitic capacitance (C_S) which includes the gate-to-source capacitance of M_1 and the gate capacitance of M_2. M_3 denotes the read access transistor which the RWL controls to read the

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Operation Table of 3T eDRAM-CIM Cell		
Input	Weight	Input × Weight
RWL	V_W	RBL
0 (V_{DD})	0/1 (V_{DD}/V_{SS})	0 (V_{DD})
1 (V_{SS})	0 (V_{DD})	0 (V_{DD})
1 (V_{SS})	1 (V_{SS})	1 (V_{SS})

Fig. 9. Schematic (left and top), operation table (left and bottom), and layout (right) of the proposed high-density 3T eDRAM-CIM cell.

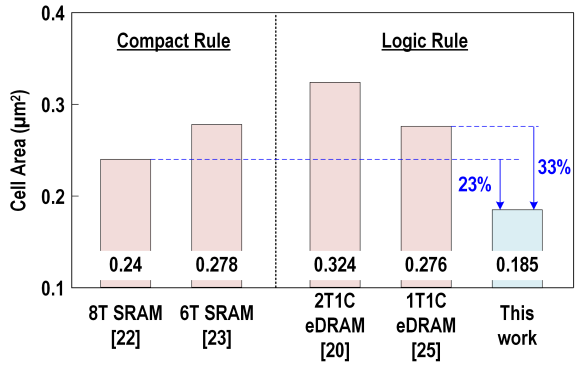


Fig. 10. Cell area comparison of the 3T eDRAM-CIM cell with state-of-the-art SRAM-CIM cells and eDRAM-CIM cells.

data to the RBL. The 3T eDRAM cell features decoupled read and write operation. Moreover, by storing the weight in the eDRAM cell and employing the RWL for transferring the IA, the 3T eDRAM cell can achieve 1b AND operation, as illustrated in the operation table. It should be noted that RBL should be pre-charged to VDD before each read or in-memory multiplication. The 3 transistors are designed with the minimum size for maximizing the cell density. The layout shows that, each 3T eDRAM-CIM cell occupies 0.185 μm^2.

Fig. 10 compares the cell area of the proposed 3T eDRAM-CIM cell with state-of-the-art (SOTA) SRAM-CIM cells and eDRAM-CIM cells. Compared with the SOTA 1T1C eDRAM-CIM cells [25] with 0.276 μm^2 based on the logic rule, the 3T eDRAM-CIM cell has 1.49× higher cell density. Even compared to the SOTA compact rule-based 8T SRAM-CIM cell [22] with 0.24 μm^2, the 3T eDRAM-CIM cell still shows 1.3× higher cell density. To support the FD-CIM, the BL switches and the related peripheral circuits for reconfiguration cause a 4.6% macro area overhead. Since the off-core data access dominates the power in a system [32], such system-level advantages gained from the FD-CIM can cover the circuit-level overhead for reconfiguration.

The retention time of the 3T eDRAM-CIM cell is ~10 μs. The worst case of needed refreshes per layer is decided by the size of the output feature in CNN models [30] and the number of tokens in Transformer-based models, respectively. For 4b operation, the 10 μs retention time would need <2 refreshes per convolutional layer for ResNet-50@ImageNet [30]. Towards ViT-B@ImageNet, the 10 μs retention time requires <1 and 0 average refreshes per matrix multiplication in each layer under 50 MHz and 400 MHz, respectively. Each refresh operation contributes <8% power overhead and 0.5% throughput reduction of the GSLP-CIM accelerator.

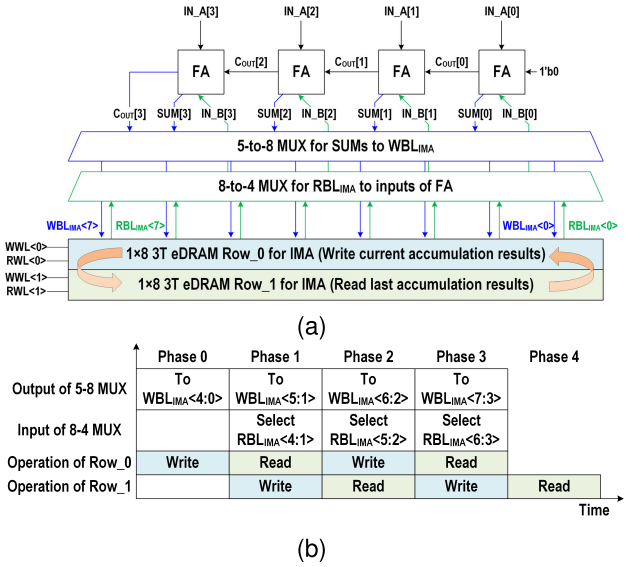


Fig. 11. (a) Schematic and (b) operation timing diagram of the proposed IMA based on ping-pong eDRAM sets.

B. In-Memory-Accumulator (IMA)

Fig. 11 takes 4b as an example to illustrate the operation principle of the proposed IMA. Compared to the conventional digital accumulator [22], which is based on the registers for storing the accumulated results, the proposed IMA employs a ping-pong 3T eDRAM set with shared BLs to save the area overhead, as shown in Fig. 11(a). Moreover, as the parasitic capacitance in each accumulation unit is reduced from ~1 fF of a register to ~0.3 fF of a BL in 28-nm CMOS technology, the power consumption can also be saved. Specifically, by reading the last accumulated results from one of the eDRAM rows for accumulating with current inputs, the current accumulated results are written into another eDRAM row. In the next phase, the role of the 2 eDRAM rows will be exchanged. As illustrated in Fig. 11(b), based on two MUXs including the 5-to-8 MUX for the accumulated results to WBLs and the 8-to-4 MUX for the RBLs to the inputs of full adders, the shift operation in the conventional digital accumulator is implemented in the IMA by selecting different BLs of the eDRAM row. Based on the post-layout simulation, the proposed IMA achieves 19.3% area and 17.4% power reduction than the conventional digital accumulator, as shown in Fig. 12.

IV. Implementation Results

Fabricated in a 28-nm CMOS technology, the proposed GSLP-CIM CNN/Transformer accelerator occupies 1.27 mm^2 active area excluding the weight/input/output buffers. Fig. 13 shows the die micrograph (top) and the layout of the SR-CIM macro (bottom). The measurement setup of the

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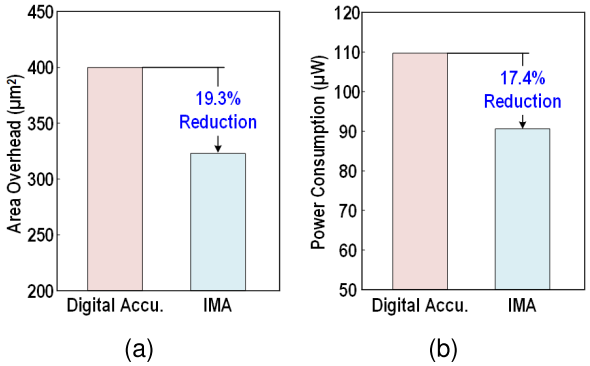


Fig. 12. (a) Area and (b) power reduction of the IMA.

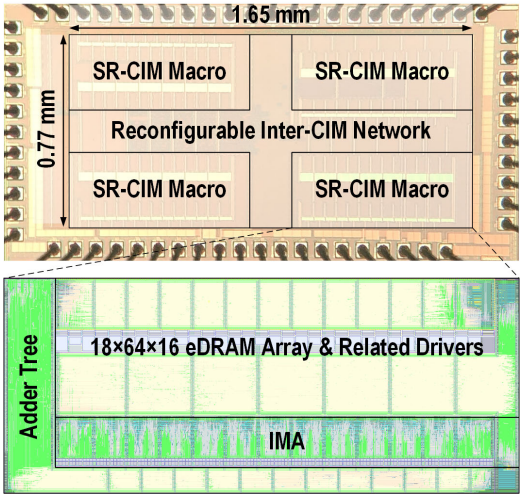


Fig. 13. Die macrograph of the proposed GSLP-CIM CNN/Transformer accelerator (top) and the layout of the SR-CIM macro (bottom).

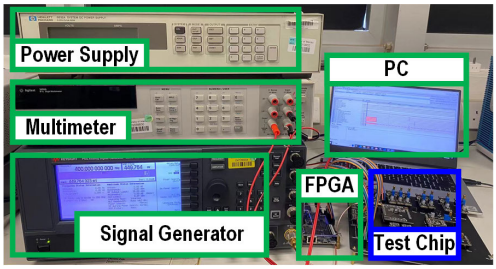


Fig. 14. Measurement setup of the GSLP-CIM CNN/Transformer accelerator.

GSLP-CIM accelerator is demonstrated in Fig. 14. A signal generator is used to generate the clock signal for the test chip. Similar to the prior works [30], [31], an FPGA is employed to generate and load the related weights and activations to the test chip. Thanks to the proposed FD-CIM, data access from the FPGA can be significantly reduced compared to the conventional WS-CIM designs [8], [17], [18], [19], [20], [21], [26], [27], [28], [31]. Similar to prior work [33], for processing the Softmax and Layer Normalization, a single instruction multiple data (SIMD) core is required. Simulation results show that such SIMD core causes ∼5% larger power consumption of the GSLP-CIM accelerator. Recent techniques [34], [35], [36] can be applied to reduce the system latency which would limited by Softmax and Layer Normalization.

Table I presents the chip summary. The GSLP-CIM accelerator with 2 × 2 SR-CIM macros contains 72-kb on-chip eDRAM. Due to the mis-designed I/O interface driver, the measured maximum operation frequency is bound to 50 MHz. This work further estimated the chip at 400 MHz based on the measurement results to demonstrate the advantage of the proposed GSLP-CIM accelerator as pre-designed and post-layout simulated. With 0.9 V supply voltage, the measured and estimated GSLP-CIM accelerator consumes 8.87 mW and 57.7 mW and performs the 4b compute energy efficiency of 2.93 TOPS/W and 3.55 TOPS/W, respectively.

TABLE I
CHIP SUMMARY

	Measured	Estimated
Technology	28nm	
Die Area (mm²)	1.96	
Core Area (mm²)	1.27	
Memory Size (Kb)	72	
Supply Voltage (V)	0.9	
Peak Frequency (MHz)	50	400
Power (mW)	8.87	57.7
4b Macro Throughput (TOPS)	0.006	0.051
4b Accelerator Throughput (TOPS)	0.026	0.205
4b Macro Throughput Density (TOPS/mm²)	0.03	0.25
4b Accelerator Throughput Density (TOPS/mm²)	0.02	0.16
4b Macro Energy Efficiency (TOPS/W) *	4.25	5.15
4b Accelerator Energy Efficiency (TOPS/W)	2.93	3.55

* Exclude the energy consumption of the configuration MUXs in the macro for flexible dataflow

Following the proposed FD scheme illustrated in Fig. 3 and the related workload mapping in Fig. 6, the CIM macro array in the GSLP-CIM accelerator can be further scaled up and exhibits superior scalability. While scaling by N, the required I/O bandwidth of conventional fully parallel weight stationary CIM [29] is scaled with N^2, and GSLP-CIM is scaled with N. The required I/O bandwidth will be significantly relaxed while scaling the CIM array with a larger ratio. Thanks to the linear increased I/O requirement of GSLP-CIM, the energy efficiency of I/O and the inter-CIM network would not increase with the scaling up of the CIM array. Measurement and Simulation results show that the energy efficiency of the I/O buffer and the reconfigurable inter-CIM network is about 0.81pJ/bit and 5.5 fJ/bit, respectively, in both 2 × 2 and 5 × 5 CIM array sizes.

Fig. 15 presents the area and power breakdown of the GSLP-CIM CNN/Transformer accelerator and the SR-CIM macro. Each SR-CIM macro occupies 16.2% area of the whole GSLP-CIM accelerator, while the reconfigurable inter-CIM communication network uses 35.2%. In each SR-CIM macro, the eDRAM array and related WL/BL drivers occupy 59.2% area, while the adder tree and IMA uses 19.4% and 21.4%, respectively. In terms of energy breakdown, each SR-CIM macro consumes 24.4% energy of the entire GSLP-CIM accelerator, while the reconfigurable inter-CIM network dissipates 2.4%. The eDRAM array and related drivers consume 41% of energy within each SR-CIM macro. The adder tree and IMA dissipate 29% and 30% energy of the SR-CIM macro, respectively.

Table II compares the performance of this work and SOTA-related works. For a better and fairer evaluation of the performances of system-level accelerators, this work employs

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TABLE II
PERFORMANCE COMPARISON WITH STATE-OF-THE-ART RELATED WORKS

		This work		JSSC'24 [26]	JSSC'24 [21]	JSSC'22 [8]	JSSC'24 [31]	JSSC'24 [18]	TCAS-I'23 [28]
Technology		28nm		65nm	65nm	40nm	28nm	65nm	65nm
MVM Implementation		Digital CIM		Digital MVM	Analog CIM	Digital MVM	Analog CIM	Analog CIM	Digital NM
Integration Level		System		System	System	System	System	Macro	Macro
Measured / Estimated / Simulated		Measured	Estimated	Measured	Measured	Measured	Measured	Measured	Simulated
Memory Cell Structure		3T eDRAM		-	6T SRAM	-	3T2C eDRAM	1T1C eDRAM	7T SRAM
Bit-cell Area (μm^2)		**0.185**		-	0.257	-	0.459	22.08	2.82
Memory Size (Kb)		4×18		-	64	-	96×100	16	80
Macro Area (mm^2)		0.206		-	0.43	-	0.103	0.57	0.473
Core Area (mm^2)		1.27		6.4	8.32	29.2	20.25	-	-
Input Precision		4b/8b		8b	1b-8b	8b	4b/8b	8b	1b-16b
Weight Precision		4b/8b		8b	2b/4b/6b/8b	8b	5b/9b	8b	1b-16b
Supply Voltage (V)		0.9		0.9-1.2	0.62-1.0	1.1	1.0	1.0-1.2	0.8-1.1
Supported Dataflow [a]		**WS, OS, RS**		WS	WS	WS	WS	WS	WS
Norm. Off-core Data Access [b]	ResNet-50 [h]	**0.042×**		1×	1×	1×	1×	1×	1×
	ViT-B [h]	**0.034×**		1×	1×	1×	1×	1×	1×
Norm. 4b [f] Peak System Throughput (IPS [c])	ResNet-50 [h]	3.1	24.5	82.1	7.91	73.7 [g]	460.9	0.38	2.56
	ViT-B [h]	0.71	5.66	18.2	1.76	16.4 [g]	97.0	0.08	0.57
Norm. 4b [f] Peak System Throughput Density (IPS/mm^2)	ResNet-50 [h]	2.41	**19.3**	12.8	0.95	2.52 [g]	22.8	0.67	5.41
	ViT-B [h]	0.56	**4.46**	2.84	0.21	0.56 [g]	4.79	0.14	1.21
Norm. 4b [f] Peak System Compute Energy Efficiency (IPW [d]) [e]	ResNet-50 [h]	349.9	**423.9**	80.1	190.2	176.2 [g]	590.8	381.3	394.3
	ViT-B [h]	80.6	**97.6**	17.8	42.2	39.1 [g]	131.1	84.6	87.5
Application		CNN, Transformer		CNN, Transformer	CNN, FC	CNN	CNN	CNN	CNN

[a] WS: Weight stationary, OS: Output stationary, RS: Row stationary; [b] Including the on-chip global buffer access and the off-chip access of weights and activations with array size normalized to 72-kb;
[c] IPS: Inferences per second = throughput / operations per inference [11]; [d] IPW: Inferences per watt = energy efficiency / operations per inference;
[e] Excluding the off-core data access energy; [f] Normalized to 4b input, 4b weight; [g] Supporting both inference and training; [h] For ImageNet classification

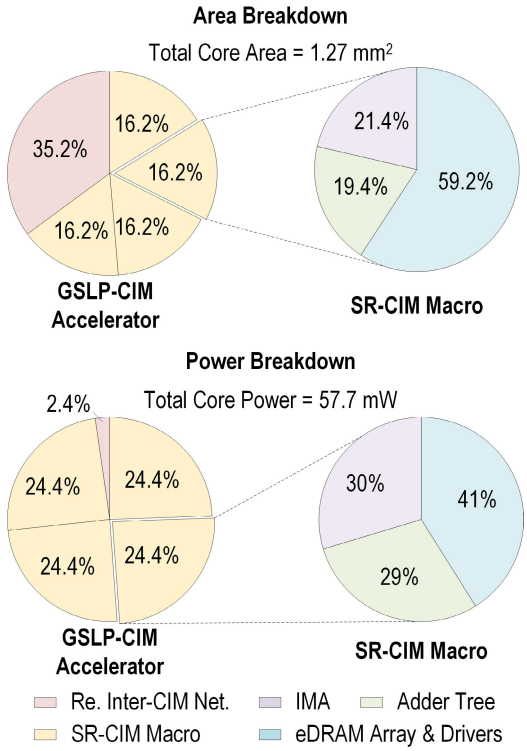


Fig. 15. Area and power breakdown of the GSLP-CIM CNN/Transformer accelerator and the SR-CIM macro.

the inferences per second (IPS) as the system throughput [11]. The IPS is calculated by the throughput over the operations per inference in specific tasks. Following the definition of IPS, this work further employs the IPS/mm^2 as the system throughput density, and the inferences per watt (IPW) as the system compute energy efficiency. The IPW can be calculated by the compute energy efficiency over the operations per inference.

Thanks to the proposed GSLP-CIM architecture with SR-CIM macro, this work can support FD including WS, OS, and RS dataflow, while SOTA works can only support the WS dataflow. The FD-CIM in this work achieves 95.8% and 96.6% off-core data access reduction for accelerating the classic CNN model (i.e., ResNet-50@ImageNet) and Transformer-based model (i.e., ViT-B@ImageNet) than SOTA works. Under the same active area as the proposed GSLP-CIM, the WS-CIM can be equipped with an estimated 500-kb global buffer. The off-chip memory access would be reduced by 8.5% and 5% for ResNet-50@ImageNet and ViT-B@ImageNet, respectively, significantly lower than the 95.8% and 96.6% achieved by the proposed GSLP-CIM.

Cooperating with the proposed high-density 3T eDRAM-CIM cell and area-efficient IMA, even excluding the off-core data access power and latency, this work achieves the system throughput density of 19.3 IPS/mm^2 and 4.46 IPS/mm^2, which is 1.51× and 1.57× higher than SOTA system level work [26] for accelerating the ResNet-50@ImageNet and ViT-B@ImageNet, respectively, and comparable to [31]. Towards the system compute energy efficiency excluding the data access energy, this work achieves 423.9 IPW and 97.6 IPW which is 2.23× and 2.31× better than SOTA system level design based on SRAM-CIM [21] for accelerating the ResNet-50@ImageNet and ViT-B@ImageNet, respectively. Even compared with SOTA macro level work [28], this work still performs 1.08×, and 1.12× higher system compute energy efficiency for accelerating the ResNet-50@ImageNet and ViT-

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B@ImageNet, respectively. A 3.9 pJ/bit of HBM2 [37] is used for off-chip memory access estimation. It shows less than 30% influence on system energy efficiency and less than 10% influence on system throughput, thanks to the FD-CIM. While the energy efficiency advantage of SOTA fixed-dataflow CIM compared to digital accelerators diminishes after taking off-chip memory access into account.

V. Conclusion

This article presented a GSLP-CIM CNN/Transformer accelerator in 28-nm CMOS. Based on the GSLP architecture with reconfigurable inter-CIM communication network and the SR-CIM macro with reconfigurable weight/input/output memory ratio, the FD is firstly supported in CIM processor for reducing >95% data access in both CNN (ResNet-50@ImageNet) and Transformer-based (ViT-B@ImageNet) models. In each eDRAM-based SR-CIM macro, the proposed high-density 3T eDRAM-CIM cell reaches 1.49× and 1.3× higher cell density than SOTA 1T1C eDRAM-CIM cell and 8T SRAM-CIM cell, respectively. Thanks to the proposed IMA based on the ping-pong eDRAM set, the area and power overhead of the digital accumulation in each eDRAM-CIM macro are reduced by 19.3% and 17.4%, respectively.

References

[1] T.-H. Hsu et al., "A 0.8 V intelligent vision sensor with tiny convolutional neural network and programmable weights using mixed-mode processing-in-sensor technique for image classification," *IEEE J. Solid-State Circuits*, vol. 58, no. 11, pp. 3266–3274, Nov. 2023.

[2] Y. Gong et al., "22.7 DL-VOPU: An energy-efficient domain-specific deep-learning-based visual object processing unit supporting multi-scale semantic feature extraction for mobile object detection/tracking applications," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2023, pp. 1–3.

[3] J. Jung, S. Kim, W. Jang, B. Seo, and K. J. Lee, "An energy-efficient, unified CNN accelerator for real-time multi-object semantic segmentation for autonomous vehicle," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 71, no. 5, pp. 2093–2104, May 2024.

[4] T. Tambe et al., "A 16-nm SoC for noise-robust speech and NLP edge AI inference with Bayesian sound source separation and attention-based DNNs," *IEEE J. Solid-State Circuits*, vol. 58, no. 2, pp. 569–581, Feb. 2023.

[5] H. Wang, Y. Zhu, H. Adam, A. Yuille, and L.-C. Chen, "MaX-DeepLab: End-to-end panoptic segmentation with mask transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 5459–5470.

[6] Y. Wang et al., "An energy-efficient transformer processor exploiting dynamic weak relevances in global attention," *IEEE J. Solid-State Circuits*, vol. 58, no. 1, pp. 227–242, Jan. 2023.

[7] A. Dosovitskiy et al., "An image is worth 16 × 16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[8] K. Prabhu et al., "CHIMERA: A 0.92-TOPS, 2.2-TOPS/W edge AI accelerator with 2-MByte on-chip foundry resistive RAM for efficient training and inference," *IEEE J. Solid-State Circuits*, vol. 57, no. 4, pp. 1013–1026, Apr. 2022.

[9] Y.-H. Chen, T. Krishna, J. Emer, and V. Sze, "Eyeriss: An energy-efficient reconfigurable accelerator for deep convolutional neural networks," *IEEE J. Solid-State Circuits*, vol. 52, no. 1, pp. 127–138, Jan. 2017.

[10] J. S. Park et al., "A multi-mode 8K-MAC HW-utilization-aware neural processing unit with a unified multi-precision datapath in 4 nm flagship mobile SoC," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2022, pp. 246–248.

[11] C.-Y. Du et al., "A 28 nm 11.2TOPS/W hardware-utilization-aware neural-network accelerator with dynamic dataflow," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2023, pp. 332–334.

[12] *Nvdia Deep Learning Accelerator*, Nvidia, Santa Clara, CA, USA, 2018. [Online]. Available: http://nvdla.org

[13] A. Parashar et al., "SCNN: An accelerator for compressed-sparse convolutional neural networks," in *Proc. ACM/IEEE 44th Annu. Int. Symp. Comput. Archit. (ISCA)*, Jun. 2017, pp. 27–40.

[14] Z. Du et al., "ShiDianNao: Shifting vision processing closer to the sensor," in *Proc. ACM/IEEE 42th Annu. Int. Symp. Comput. Archit. (ISCA)*, Jun. 2015, pp. 92–104.

[15] H. Kwon, P. Chatarasi, V. Sarkar, T. Krishna, M. Pellauer, and A. Parashar, "MAESTRO: A data-centric approach to understand reuse, performance, and hardware cost of DNN mappings," *IEEE Micro*, vol. 40, no. 3, pp. 20–29, May 2020.

[16] C.-J. Jhang, C.-X. Xue, J.-M. Hung, F.-C. Chang, and M.-F. Chang, "Challenges and trends of SRAM-based computing-in-memory for AI edge devices," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 68, no. 5, pp. 1773–1786, May 2021.

[17] C.-T. Lin et al., "DIMCA: An area-efficient digital in-memory computing macro featuring approximate arithmetic hardware in 28 nm," *IEEE J. Solid-State Circuits*, vol. 59, no. 3, pp. 960–971, Mar. 2024.

[18] S. Xie, C. Ni, A. Sayal, P. Jain, F. Hamzaoglu, and J. P. Kulkarni, "EDRAM-CIM: Reconfigurable charge domain compute-in-memory design with embedded dynamic random access memory array realizing adaptive data converters," *IEEE J. Solid-State Circuits*, vol. 59, no. 6, pp. 1950–1961, Jun. 2024.

[19] C. Yu, T. Yoo, H. Kim, T. Kim, K. Chai, and B. Kim, "A logic-compatible eDRAM compute-in-memory with embedded ADCs for processing neural networks," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 68, no. 2, pp. 667–679, Feb. 2021.

[20] S. Ha, S. Kim, D. Han, S. Um, and H.-J. Yoo, "A 36.2 dB high SNR and PVT/leakage-robust eDRAM computing-in-memory macro with segmented BL and reference cell array," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 69, no. 5, pp. 2433–2437, May 2022.

[21] J. Yue et al., "An energy-efficient computing-in-memory NN processor with set-associate blockwise sparsity and ping-pong weight update," *IEEE J. Solid-State Circuits*, vol. 59, no. 5, pp. 1612–1627, May 2024.

[22] Y. He et al., "A 28nm 38-to-102-TOPS/W 8b multiply-less approximate digital SRAM compute-in-memory macro for neural-network inference," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2023, pp. 130–132.

[23] B. Wang et al., "A 28 nm horizontal-weight-shift and vertical-feature-shift-based separate-WL 6T-SRAM computation-in-memory unit-macro for edge depthwise neural-networks," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, 2023, pp. 134–136.

[24] S. Kim et al., "Scaling-CIM: EDRAM in-memory-computing accelerator with dynamic-scaling ADC and adaptive analog operation," *IEEE J. Solid-State Circuits*, vol. 59, no. 8, pp. 2694–2705, Aug. 2024.

[25] S. Kim et al., "A reconfigurable 1T1C eDRAM-based spiking neural network computing-in-memory processor for high system-level efficiency," in *Proc. IEEE Int. Symp. Circuits Syst. (ISCAS)*, May 2023, pp. 1–5.

[26] S. S. Sundaram, Y. Khodke, Y. Li, S.-J. Jang, S.-S. Lee, and M. Kang, "FreFlex: A high-performance processor for convolution and attention computations via sparsity-adaptive dynamic frequency boosting," *IEEE J. Solid-State Circuits*, vol. 59, no. 3, pp. 855–866, Mar. 2024.

[27] C.-Y. Yao, T.-Y. Wu, H.-C. Liang, Y.-K. Chen, and T.-T. Liu, "A fully bit-flexible computation in memory macro using multi-functional computing bit cell and embedded input sparsity sensing," *IEEE J. Solid-State Circuits*, vol. 58, no. 5, pp. 1487–1495, May 2023.

[28] H. Kim, J. Mu, C. Yu, T. T. Kim, and B. Kim, "A 1–16b reconfigurable 80Kb 7T SRAM-based digital near-memory computing macro for processing neural networks," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 70, no. 4, pp. 1580–1590, Apr. 2023.

[29] H. Jia et al., "Scalable and programmable neural network inference accelerator based on in-memory computing," *IEEE J. Solid-State Circuits*, vol. 57, no. 1, pp. 198–211, Jan. 2022.

[30] Y. Zhan, W.-H. Yu, K.-F. Un, R. P. Martins, and P.-I. Mak, "A 28-nm 18.7 TOPS/mm^2 89.4-to-234.6 TOPS/W 8b single-finger eDRAM compute-in-memory macro with bit-wise sparsity aware and kernel-wise weight update/refresh," *IEEE J. Solid-State Circuits*, vol. 59, no. 11, pp. 3866–3876, Nov. 2024.

[31] S. Kim et al., "DynaPlasia: An eDRAM in-memory computing-based reconfigurable spatial accelerator with triple-mode cell," *IEEE J. Solid-State Circuits*, vol. 59, no. 1, pp. 102–115, Jan. 2024.

[32] V. Sze, Y.-H. Chen, T.-J. Yang, and J. S. Emer, "Efficient processing of deep neural networks: A tutorial and survey," *Proc. IEEE*, vol. 105, no. 12, pp. 2295–2329, Dec. 2017.

Authorized licensed use limited to: University of Shanghai For Science and Technology. Downloaded on September 29,2026 at 13:33:24 UTC from IEEE Xplore. Restrictions apply.

[33] F. Tu et al., "TranCIM: Full-digital bitline-transpose CIM-based sparse transformer accelerator with pipeline/parallel reconfigurable modes," *IEEE J. Solid-State Circuits*, vol. 58, no. 6, pp. 1798–1809, Jun. 2023.
[34] J. R. Stevens, R. Venkatesan, S. Dai, B. Khailany, and A. Raghunathan, "Softermax: Hardware/software co-design of an efficient softmax for transformers," in *Proc. ACM/IEEE Design Auto. Conf. (DAC)*, Dec. 2021, pp. 469–474.
[35] K. Hong et al., "FlashDecoding++: Faster large language model inference on GPUs," 2023, *arXiv:2311.01282*.
[36] J. Shah, G. Bikshandi, Y. Zhang, V. Thakkar, P. Ramani, and T. Dao, "FlashAttention-3: Fast and accurate attention with asynchrony and low-precision," 2024, *arXiv:2407.08608*.
[37] M. O'Connor et al., "Fine-grained DRAM: Energy-efficient DRAM for extreme bandwidth systems," in *Proc. 50th Annu. IEEE/ACM Int. Symp. Microarchit. (MICRO)*, Oct. 2017, pp. 41–54.
[38] A. Guo et al., "A 22 nm 64kb lightning-like hybrid computing-in-memory macro with a compressed adder tree and analog-storage quantizers for transformer and CNNs," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2024, pp. 570–572.
[39] K. Yoshioka, "A 818–4094TOPS/W capacitor-reconfigured CIM macro for unified acceleration of CNNs and transformers," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2024, pp. 574–576.
[40] S. Liu, H. Zhu, C. Chen, L. Zhang, and C.-J. R. Shi, "XNORAM: An efficient computing-in-memory architecture for binary convolutional neural networks with flexible dataflow mapping," in *Proc. 2nd IEEE Int. Conf. Artif. Intell. Circuits Syst. (AICAS)*, Aug. 2020, pp. 21–25.
[41] X. Chen et al., "A systolic computing-in-memory array based accelerator with predictive early activation for spatiotemporal convolutions," in *Proc. IEEE 5th Int. Conf. Artif. Intell. Circuits Syst. (AICAS)*, Jun. 2023, pp. 1–5.



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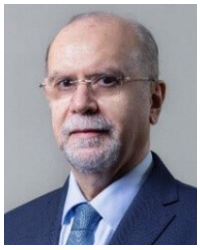
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